

"Like, Are You Okay With Being Recorded?": Understanding and Designing for Consent to Recording and Livestreaming

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Abstract

Unwanted streaming and recording have been extensively examined in HCI and privacy research, raising persistent concerns about surveillance and privacy. While numerous interventions have been proposed to mitigate these harms, few center consent as the primary framework for understanding and addressing them. This paper presents a speculative design study of VR dating with 16 stakeholders identifying as women and/or LGBTQIA+ (demographics at elevated risk of nonconsensual acts) to elucidate approaches to designing consent mechanics for video recording, which we define as technologies for giving, receiving, and denying permission to be captured in a live video stream or recording. Findings surfaced three core design goals: Visual interruptions to initiate explicit consent exchange, consent exchange to define private versus public social contexts, and platform-level defaults and user-controlled blocking tools to withhold and revoke consent.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

Keywords

Consent; Livestreaming; Unwanted Recording; VR Dating; Social VR

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1 Introduction

Prior HCI work has studied consent (i.e., voluntary agreement) as a contemporary lens for understanding and mitigating harm in computer-mediated communication [20]. Under a consent lens, the notion of objectively harmful behavior is rejected, with harm instead being subjectively defined on a person-by-person basis as unwanted behavior. A consent lens enables preventative—rather than reactive—solutions to harm through the scaffolding of consent exchange: how people give, receive, and deny agreement to a given behavior. The lens of consent has been applied across various HCI contexts where harmful and unwanted interactions may occur, including intimate sexual activity [4], mobile social media apps [20], social VR [28], and human-robot interactions [30].

Another context that could benefit from a consent lens is livestreaming and recording, given the prevalence of streaming platforms like Twitch and YouTube, and well-documented recording-based harms [5, 10, 21, 24, 36, 37] such as unwanted disclosure of personal information and nonconsensual distribution of sexual videos (e.g., revenge porn). Participants in prior research on livestreaming have even directly requested “stronger notifications of nearby livestreaming and the ability to consent to, and/or to opt out of, taking part in the broadcast” [14].

While HCI literature has addressed consent to recording [21, 36, 37] and associated privacy concerns [14, 21, 36, 37], consent *mechanics* for livestreaming have yet to be developed. In this paper, we present a speculative design study of VR dating to elucidate approaches to designing consent mechanics for video recording, which we define as technologies for giving, receiving, and denying permission to be captured in a live video stream or recording.

We use VR dating as a speculative context because it brings together many known sites of problematic streaming and recording, including social VR platforms [19, 24], AR glasses in physical-world environments [1, 12], and intimate sexual interactions, both in VR and in the physical world, that could lead to revenge porn distribution [2, 5]. Our research question is: *How can AR/VR platforms design for consent to recording and livestreaming?*

Through group-based design workshops with 16 women and LGBTQIA+ stakeholders, we leveraged their prior experience with mobile dating apps and social VR to elucidate three design approaches for consent to recording: visual interruptions to initiate

explicit consent exchange, consent exchange to define private versus public social contexts, and platform-level defaults and user-controlled blocking tools to withhold and revoke consent.

2 Related Work

2.1 Proliferation of, and Problematic Practices With, Streaming and Recording

Livestreaming and digital recording technologies have rapidly evolved from niche tools to mainstream features embedded in daily life. Today, people use platforms such as Twitch, TikTok Live, and YouTube Live to broadcast themselves in real time, often from smartphones [3]. These platforms serve diverse purposes, from entertainment to education [31, 35]. At the same time, recording tools are increasingly integrated into digital platforms and devices, enabling users to capture audio and video across various contexts—including through built-in screen recording features that can be activated without in-app notifications [9, 23, 33]. Looking ahead, speculative and emerging technologies suggest even broader proliferation of livestreaming and recording, such as livestreaming of social Virtual Reality (VR) to outside audiences [19] and Augmented Reality (AR) glasses for hands-free video capture.

Recording in livestreams poses a multitude of privacy concerns, such as real-time, unintended public exposure of personal information [21, 37]. Streamers are also concerned about bystanders' privacy, particularly roommates and family members who may appear in the stream without consent [36]—research shows that bystanders may not be able to reliably tell when they are being streamed [14]. Accidental broadcasting of background audio is a related challenge, especially when the audio includes discussion of private personal issues [36]. Social VR environments introduce further issues, such as covert and unwanted recording of other users [10, 24]. Relatedly, concerns about nonconsensual recording through AR glasses have been found [1], especially due to their subtle, always-on nature [12, 18].

Harms associated with unwanted or nonconsensual sharing of sexual recordings and imagery (e.g., revenge porn) are also well researched. Revenge porn has been linked with severe mental health impacts, such as PTSD and depression [5]. Studies highlight barriers to help-seeking like stigma and shame, emphasizing the need for inclusive, victim-centered support systems [6, 22] and consent-aware design [2, 17]. Relatedly, nonconsensual image sharing has become more prevalent, which may be exacerbated by the use of generative AI to create deepfake pornography [34]. Other work with creators of intimate content documents the various threats and harms they encounter when consensually sharing content [26], as well as strategies to prevent accidental disclosure of sensitive images [16].

2.2 Solutions for Streaming and Recording Privacy

To manage privacy concerns, streamers often use technical and behavioral strategies such as overlays to hide parts of the screen and muting or delaying streams. Some also try to interpret non-verbal cues (e.g., someone avoiding the camera) or take proactive steps like asking for pre-approval, but these efforts are inconsistent

and easily misinterpreted [21, 36, 37]. Beyond user strategies, one promising approach to improving situational awareness in immersive environments is "Vice VRsa", a prototype system developed by Do et al. [13]. It explores a set of real-time awareness mechanisms meant for both users and bystanders, such as visual indicators inside the VR headset to alert users of nearby bystanders, and external cues (like LED signals) to inform bystanders whether recording or passthrough is active. These bidirectional signals aim to scaffold mutual awareness and support situational negotiation of privacy boundaries [13].

2.3 Opportunity for Streaming and Recording Consent Mechanics

Across studies, participants have consistently expressed a desire for greater control over being recorded or streamed. In mobile livestreaming contexts, nearly all interviewees wanted clearer notifications and the ability to consent or opt out [14]. Similarly, bystanders observing AR glasses preferred being asked for consent and supported technologies that could block unauthorized capture [12]. The HCI community has also increasingly used consent as a design lens to address harm and harassment, and support user agency. In social VR, it helps to define and prevent interpersonal harm [28, 38], while other scholars have extended feminist models of affirmative and embodied consent to emerging technologies [20, 30]. This work positions consent as a tool for designing safer digital interactions [39], including our own efforts to understand and design for consent to livestreaming and recording.

3 Methodology

We investigated our research questions through a series of group-based design workshops that blended elements of focus group discussion and participatory design. These were held in the Midwestern United States with 16 prospective VR dating users who self-identify as women and/or LGBTQIA+. This study was approved by our university's institutional review board (IRB). We drew insights from a prior study that developed methodological guidance for participatory design of emerging technologies for sexual violence, which shaped both our recruitment approach and methodological choices [11]. We also sought feedback from a certified sexual assault nurse examiner as well as psychology researchers with expertise in sexual violence. Across several workshop sessions, participants reflected on their personally preferred consent practices within VR dating and constructed scenarios of consensual interactions, during which they voluntarily identified recording as a significant area of focus for their scenarios.

We sought participants who identified as women and/or LGBTQIA+ because those demographics are predominantly survivors of non-consensual acts in-person [15] and online [25, 27, 29], thus positioning them to most benefit from this research. We further sought prior experience with either dating apps or social VR platforms and an interest in using VR for dating (personal use with VR dating platforms was not required because of their limited availability to the public at the time of study). Recruitment methods included messages on social media, mailing lists for a university's students and sororities at that university, as well as snowball sampling. Of the 16 participants, 14 identified as women and 2 as non-binary. Seven

were heterosexual, 4 were bisexual, 2 were homosexual, one was pansexual, and two chose not to disclose their sexual orientation. Participants were compensated with a \$50 gift card.

The 16 participants were organized into four groups of three to five individuals, and each group took part in a series of up to three sessions lasting three hours each. This paper reports on the analysis of two of the three study sessions. During the first session, participants individually identified potential nonconsensual interactions that might occur within VR dating platforms. These individual reflections were followed by group discussions, where participants shared their personally preferred or practiced approaches to consent that could address the scenarios they identified. In the second session, participants collaboratively designed consent mechanisms to mitigate the nonconsensual interactions discussed in the first session. The third session is beyond the scope of this paper.

Transcripts of the design sessions, along with visual artifacts produced by participants, were subjected to reflexive thematic analysis [7]. Three researchers began by individually reviewing the data and taking informal notes to build familiarity. This was followed by collaborative coding and the development of an initial thematic map, which was iteratively refined through ongoing group discussions and critical reflection. When discussing their preferred consent practices for certain interactions, many participants centered their discussion around scenarios involving recording, hence the focus of this paper on elucidating approaches to designing consent mechanics for video recording. The writing of this manuscript thus served as the final stage of reflexive thematic analysis pertaining to our investigation of the aforementioned research questions.

4 Findings

There was ample discussion among participants about consent to recording in virtual and augmented reality dating situations due to competing reasons for why they would and would not want their interactions to be recorded. They imagined VR dating environments being used for incredibly intimate sexual interactions that would be subject to potential recording, including “VR strip clubs” (P12) and “kink-specific spaces” (P14). For such interactions, recording could be used simultaneously for safety (by having evidence of sexual violence or harassment) and for harm (through misuse of the recordings, such as for revenge porn). For example, P10 speculated that consenting to the recording of virtual sexual acts could enable automated reporting of sexual harassment:

“They’re automatically reported, the developers then have a recording of the instance, like being harassed has, or like some sort of a note within the developers log. So then, that way, it can spare the user from having to deal with all that burden of like having to report and having to like, advocate for themselves, because that can be hard to me, as well.” (P10)

With these types of intimate interactions as a backdrop, participants articulated three interconnected design goals for scaffolding consent to recording and streaming within VR dating environments and during physical-world dates with AR glasses: 1) utilizing the visual affordances of VR/AR to purposely interrupt interaction with explicit consent prompts for recording or livestreaming; 2) embedding consent exchanges at key transition points to define private versus public social contexts; and 3) establishing platform-level

defaults and user-controlled tools to block or prevent recording when consent is withheld or revoked.

4.1 Design Goal 1: Visual Interruptions to Initiate Explicit Consent Exchange

Participants emphasized that VR’s “assumed anonymity” (P6)—where avatars can be anything from animals to abstract shapes—and its sense of being “not real” (P16) encourage users to treat recording as trivial and disconnected from accountability or “real” consequences. In response to this understanding, participants proposed leveraging the unique affordances of immersive and embodied technologies to institute explicit, hard-to-miss consent exchanges that require mutual agreement before any recording or streaming can occur. They felt that such exchanges would reduce unintentional cases of non-consensual recording by normalizing the exchange of consent and making each partner’s boundaries around recording immediately clear.

To achieve this, some participants envisioned platform-driven consent prompts that appear directly within users’ fields of vision, pausing the experience until both parties actively affirm or deny permission to record. As P1 observed, this could mirror Zoom’s on-screen notification—“This session is being recorded”—but reimagined for immersive spaces.

These and similar suggestions emphasize two core benefits. First, by making consent exchange a visually obvious event, platforms can help to reduce the chance that such prompts are missed while also compelling users to reflect on the seriousness of the exchange and the resulting consensual decision. Second, these suggestions foreground a platform-driven approach to consent exchange, wherein platforms themselves are tasked with initiating these explicit exchanges. In doing so, platforms help to reduce the possibility of consent exchanges not occurring due to conceptions about permission being optional or unnecessary.

4.2 Design Goal 2: Consent Exchange to Define Private Versus Public Social Contexts

Participants frequently pointed to the ambiguity of private space in virtual environments as a core source of anxiety: they were not sure whether entering a new room, call, or virtual area meant that recording could or could not occur, nor were they sure how others would interpret these boundaries. When faced with this uncertainty, participants suggested that consent exchange should occur before entering private rooms or traversing into spaces where private interactions or sensitive discussions could be reasonably expected. For example, P1 described: “Like, right before you enter the room, like, are you okay with being recorded? Yes or no?”

In this way, there is no longer a lack of clarity about what constitutes “private,” because rather than basing such a definition upon social norms or platform-defined labels, participants imagined mutual agreements between users, through exchanging consent to recording, being the primary mechanism for defining privacy.

Although many suggestions frame these exchanges as being initiated by the platform itself (per discussion relating to Section 4.1), and therefore requiring some sort of reliance on platform-set expectations, this does not necessarily need to be the case. For example, these prompts could be given in regular intervals or before

any transition across spaces, or they could be provided as tools for users to activate themselves at any point. However implemented, these suggestions framed consent as not only boundary-setting, but also context-building, creating a shared sense of what should be expected from any given space or interaction.

4.3 Design Goal 3: Platform-Level Defaults and User-Controlled Blocking Tools to Withhold and Revoke Consent

Participants also stressed that consent is not a one-time act. They emphasized that people need to be able to refuse or revoke consent at any point during or after an interaction, especially as their sense of safety or comfort changes. For this purpose, blocking—which is often treated as a social or safety feature—was reimagined by participants as a consent enforcement tool: a way to halt or undo recording permissions. Participants imagined features that allowed users to prevent recording or livestreaming at any time.

For instance, P7 described a virtual room toggle that both parties must “unlock” before recording or screen-sharing becomes possible—and can relock at any time to prevent these features: “...*both parties can [...] turn the switch on and off to [...] lock the room.*” P1 extended this idea to a vicinity-based system, where anyone approaching a nonconsenting user would see their recording interface “go blank, [...] it’ll just be [...] a black or white screen.”

Participants felt that these features would grant users complete autonomy over their consent, respecting changing comfort levels and unanticipated interactions. Also important was that these features be on by default. In other words, participants wanted systems where non-consent to recording would be the default option, and where users could “opt-in” to being recorded through mutual and ongoing agreement, rather than being forced to “opt-out” and face the potential of being recorded nonconsensually before they do so.

5 Discussion and Conclusion

The design goals discovered through our study shift the framing of video recording from an implicit privacy problem to an explicit consent negotiation process. In our VR dating context—and more broadly in any recording-capable medium (from social VR and AR wearables to mobile apps and livestreams)—existing research has documented how bystanders and even streamers often suffer unexpected exposures with little recourse [14, 21, 36, 37]. Prior HCI research into recording typically describes these as privacy violations [21, 36, 37]. In contrast, we treat consent itself as the organizing concept, as suggested in other work on computer-mediated harm and sexual activity [20, 30, 30, 38, 39]. Below, we consider each design goal, linking to prior literature on livestreaming and recording, and highlight how a consent-centered lens opens new design possibilities.

Visual Consent Mechanics: Participants advocated for explicit visual consent exchanges. This approach reflects a broader trend in HCI research toward visual indicators for recording status in livestreaming and AR, where badges, lights, and notifications signal capture [12, 14]. However, most prior prototypes focus solely on notifying users that recording is active, rather than using those same visuals to structure consent negotiation to recording—a gap noted in work on cross-reality safety and alternatives to affirmative

consent [8, 32]. To fill this gap, future research could co-design and test these visual consent mechanics with diverse stakeholders, identifying the prompts, visuals, and interfaces that best foster clear, reciprocal consent.

User-Defined and Mutually Maintained Private Spaces: Participants underscored a fundamental need for clear privacy rules in virtual and mixed-reality environments—an expectation long supported in livestreaming and recording research, where privacy concerns dominate the conversation [1, 12, 13, 21, 36, 37]. Traditional approaches tend to rely on inferring privacy based on contextual or spatial features, such as assuming that a room with limited access should be private, and designing systems that enforce behavioral norms accordingly [13]. In contrast, our participants proposed that privacy itself be established through a preemptive consent exchange. This shift opens new design avenues: rather than focusing on enforcing privacy within a defined space, future work can foreground consent negotiation as a potential tool for creating private spaces that are user-defined and mutually maintained.

Tools for Revoking and Enforcing Consent to Recording: Participants insisted on direct, revocable control over recording and livestreaming using features for blocking such technologies. This desire directly echoes findings from mobile and AR livestreaming research, where users similarly expressed a desire for direct control over being recorded and streamed in seeking to maintain personal privacy [1, 12, 14]. However, there are limited design suggestions for how this should be accomplished. Future work could leverage immersive technologies with VR/AR users and developers to prototype and evaluate blocking interfaces that could serve as tools for revoking and enforcing consent to recording.

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